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M1 Project - Bias in Negotiations with Generative AI

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1. Introduction

1.1. General Context

The integration of Generative AI, particularly Large Language Models (LLMs), into negotiation contexts represents an emerging frontier at the intersection of human-computer interaction, behavioral economics, and algorithmic fairness. The current state of the art reveals that while LLMs demonstrate remarkable capability in structuring negotiations and generating coherent strategies, they systematically exhibit, replicate, and sometimes amplify human-like cognitive biases and societal prejudices. This synthesis consolidates findings from key research papers to define the contemporary landscape, identifying established knowledge, methodological approaches, and critical gaps.

1.2. Problem Statement

As large language models (LLMs) are increasingly deployed to represent human interests in negotiation settings, critical questions arise regarding their reliability, fairness, and strategic behavior. It is unclear how these models perform in interactive, high-stakes scenarios where persuasion, compromise, and strategic reasoning are essential. Potential issues such as hidden biases, sensitivity to prompt phrasing, susceptibility to asymmetric information, and unpredictable interactions with different counterpart strategies could lead to suboptimal or unfair outcomes. Without systematic investigation, deploying LLMs as negotiation agents may risk amplifying unknown behavioral flaws, which may compromise trust and effectiveness in professional, economic, and social contexts.

1.3. Project Objectives

This project aims to develop an experimental framework to systematically investigate negotiation behaviors and biases in large language models. The objectives are as follows:

- Design Negotiation Experiments: Identify and define relevant negotiation frameworks to simulate using LLMs (e.g., bilateral bargaining)
- Develop a Technical Pipeline: Build a system using Python and commercial LLM APIs (e.g., GPT-4) to automate multi-turn negotiations between agents, manage conversational state, and enforce structured response formats.
- Create a Data Management System: Design and implement a data structure (e.g., a structured database or dataset) to store, compare, and analyze negotiation dialogues, outcomes, and model metadata across varied experimental conditions.
- Formulate and Test Hypotheses: Propose and investigate specific research questions, such as:
    - How do model biases (e.g., towards specific outcomes or phrasing) manifest in negotiations?

    - How does prompt engineering (e.g., varying agent personas, instructions, or constraints) influence negotiation strategies and outcomes?
    - Are LLMs sensitive to information asymmetry or different counterpart strategies (e.g., aggressive vs. cooperative)?
- Analyze and Document Findings: Perform comparative analysis on the collected data to characterize emergent behaviors, identify potential biases, and document unexpected strategic patterns. The final deliverable will summarize insights about LLM behavior in negotiation scenarios.

2. State of the Art

2.1. Negotiation: Foundations, Strategies, and Key Concepts

2.1.1. Definitions and Core Negotiation Concepts (BATNA, ZOPA, Concessions)

Negotiation can be defined as an interaction between two or more parties seeking to reach an agreement while pursuing conflicting interests [2, 3]. It is a decision-making process in which parties evaluate offers based on preferences, expectations, and available alternatives, often requiring significant reasoning capabilities [15]. Three concepts are central to negotiation theory. The Best Alternative to a Negotiated Agreement (BATNA) refers to the option available if no agreement is reached and defines the minimum acceptable outcome. The Zone of Possible Agreement (ZOPA) is the range of outcomes acceptable to both parties based on their BATNAs [9]. Negotiations evolve through concessions, as parties gradually adjust their positions to move closer to agreement. The size and timing of concessions signal strategy and influence perceptions of fairness and cooperation [10]. In price-based negotiations, concessions typically appear as progressive price adjustments over several rounds.

2.1.2. Decision-Making Processes in Negotiation

Negotiation decisions result from both rational evaluation and cognitive limits. Classical economic models assume rational agents seeking to maximize utility, but research in behavioral economics shows that human decision-making often departs from this ideal [13]. Negotiators rely on heuristics and reference points when assessing offers [1]. Initial proposals, perceived fairness, time pressure, and framing strongly influence outcomes, leading to bounded rationality, where decisions aim to be acceptable rather than optimal [16]. Specifically, the anchoring effect demonstrates how initial values bias subsequent judgments, a phenomenon recently observed in both human and AI agents [7, 8, 12].

2.1.3. Human–Machine Negotiation: An Emerging Challenge

The growing use of artificial intelligence represents a major change in negotiation practices, raising questions about whether AI acts as a collaborator or a competitor [2]. Large Language Models (LLMs) can generate arguments and manage multi-round exchanges, enabling human–machine negotiation [3]. While AI systems lack emotions, their behavior depends on training data and prompts.

This raises questions about trust and predictability, highlighting the need for controlled experiments comparing human and AI behavior under the same conditions.

2.1.4. Industrial Negotiations and Price-Based Asset Transactions

Industrial negotiations, such as the sale of factories or production sites, are a specific form of price-based negotiation characterized by high financial stakes and clear valuation benchmarks. Outcomes are often assessed using objective reference points such as market estimates or comparable transactions.

2.2. Cognitive Biases in Human Negotiation

2.2.1. Anchoring Bias in Negotiation

Anchoring bias refers to the tendency to rely strongly on the first information received when making decisions. In negotiation, initial offers often act as anchors that influence subsequent proposals. Early price offers shape the perceived range of acceptable outcomes and affect concession patterns. This effect is particularly strong in multi-round negotiations [7, 8, 12].

2.2.2. Framing Effects and Linguistic Biases

Framing effects occur when different presentations of the same information lead to different decisions [2, 11]. In negotiation, how an offer is formulated can significantly affect acceptance [2]. Language and context influence negotiation outcomes by focusing attention on specific aspects of the exchange [5], showing that decisions are not driven solely by objective values [14].

2.2.3. Fairness Bias and Perceptions of Justice

Negotiators often assess offers based on perceived fairness in addition to economic value. Fairness bias can lead parties to reject profitable offers if they are seen as unjust or to accept less favorable outcomes to preserve equity [10]. This bias is especially relevant in human–machine negotiations, where humans may expect AI to act fairly [4].

2.2.4. Other Relevant Cognitive Biases in Negotiation

Other cognitive biases also affect negotiation behavior, including confirmation, authority, and overconfidence [1, 14]. Prosocial tendencies may further encourage cooperation even in competitive settings [2]. Overall, these biases show that negotiation is shaped by psychological factors rather than purely rational calculation [1, 13].

2.3. Large Language Models: Mechanisms, Biases, and Behavioral Tendencies

2.3.1. How Large Language Models Operate: Prompting, Generation, and Emergent Behavior

Large Language Models (LLMs) are AI systems trained on vast data to generate human-like language. They operate by probabilistic prediction: given a prompt, the model generates responses token by token based on learned patterns. LLMs do not have intentions, beliefs, or goals, but in interactive contexts such as negotiations, they can display behaviors resembling strategic reasoning [3]. Prompting is crucial: small changes in instructions, tone, or constraints can produce very different outputs. Furthermore, the performance and computational cost of these models are directly linked to the complexity of the negotiation task [15].

2.3.2. Known Biases in Large Language Models

LLMs are not neutral. Studies show they can reproduce and amplify biases present in training data, including social, cultural, or decision-making biases [11, 14]. In negotiation contexts, LLMs exhibit human-like biases such as anchoring, framing sensitivity, and fairness tendencies [1]. These biases stem from statistical correlations, not conscious preferences, raising concerns for decision-critical applications like price negotiation [8, 12].

2.3.3. Behavioral Variability Across Models

LLMs vary across architectures, datasets, and alignment strategies, leading to distinct behavioral profiles [3, 9]. Even within one model, stochastic generation can produce different responses to identical prompts [3]. This variability complicates evaluation of consistency and fairness [14] and justifies repeated experiments and structured data collection to identify systematic patterns [9, 15].

2.4. AI and Negotiation: Existing Work and Practical Applications

2.4.1. LLMs as Negotiation Advisors and Agents

Research has studied LLMs both as advisors and active participants in negotiations. As advisors, they provide guidance, strategies, and suggestions to human negotiators [4, 6]. As active agents, LLMs can interact directly with humans or other AIs, making offers, responding to counteroffers, and adapting strategies in autonomous competitions [9]. Their style depends heavily on prompt design and can be inconsistent or unpredictable.

2.4.2. Biases in AI Negotiation

Studies show that LLMs can display human-like cognitive biases, including anchoring, framing effects, and fairness based reasoning. Even when instructed to act rationally, models may follow patterns learned from training data rather than making optimal decisions, often resulting in an "illusion of rationality" [13]. Additionally, cultural biases can affect how these models communicate

with human counterparts [5]. Controlled experiments are therefore needed to compare human and AI negotiation reliably.

2.4.3. Real-World Applications and Case Studies

AI negotiation is being tested in online marketplaces, customer service, real estate, and industrial procurement [2, 9]. Systems can automate price or contract discussions, improving efficiency [1], but practical challenges remain, such as trust, transparency, and user behavior changes when negotiating with AI [4, 18]. Industrial scenarios, like asset valuation and pricing negotiations, are useful for studying AI behavior [1, 8].

2.5. Synthesis: Limitations and Implications for Our Research

2.5.1. Scientific and Technical Limitations

Existing research shows that negotiation behavior, human or AI-driven, is shaped by cognitive and contextual factors. Many studies use simplified or artificial settings, limiting their relevance to real-world, high-stakes negotiations. Research often focuses on single biases or isolated decisions, making it hard to understand how multiple biases interact over multi-round negotiations. Different LLMs behave differently under the same conditions, and a single model can give inconsistent responses [15].

2.5.2. Practical and Ethical Considerations

Using AI negotiators in real world settings raises practical and ethical concerns regarding transparency and accountability [17]. Biases or inconsistencies can produce unfair outcomes, reduce trust, or cause economic inefficiencies [16]. Users may act more aggressively when negotiating with AI [4]. Transparency is a key question: who is responsible for AI-generated decisions, and should users always know they are negotiating with a machine?

2.5.3. Justification for Our Industrial Scenario

To address these limitations, this project uses a factory price negotiation scenario. It is realistic yet controlled, with clear BATNAs, valuation benchmarks, and a limited 5-round structure. This setup allows direct comparison of human and AI negotiation behavior, highlights potential biases such as anchoring [7] or fairness [10], and studies multi-round interactions with economically meaningful outcomes, specifically checking if ChatGPT mirrors human biased decision-making [1].

3. Proposed Strategy and Feasibility

3.1 Methodology

This section details the experimental framework, technical implementation, and analytical procedures designed to investigate bias in human-AI negotiation. The methodology employs a web-based interactive platform to collect behavioral data from human participants negotiating with a controlled Large Language Model (LLM) agent.

3.2 Experimental Design and Protocol

3.2.1 Negotiation Scenario

A standardized, single-issue distributive negotiation scenario was developed to ensure consistency and measurability. The scenario centers on the sale of a factory.

- Roles and Confidential Information: Participants are assigned the role of the Buyer (aiming to purchase at the lowest price)
- Zone of Possible Agreement (ZOPA): The overlap between RPs ($850k - $1,150k) creates a $300,000 bargaining zone. Any final agreement within this range represents a mutually beneficial outcome.
- Background Context: Both parties receive identical background information on the factory (e.g., age, location, general market conditions) to establish a common knowledge base, but the RPs are private and never disclosed.

3.2.2 Interaction Structure

Each negotiation session follows a strict turn-based protocol:

1. Initiation: The human participant sends the first message.
2. Dialogue: The AI responds, and the exchange continues for a maximum of five full conversational turns (one turn = one human message + one AI response).
3. Final Decision: After the fifth AI response, the human submits a final offer. The AI agent then evaluates this offer against its programmed decision logic and outputs a final Accept or Decline decision, terminating the negotiation.

3.2.3 Participant Recruitment and Demographics

- Target Sample: 80-100 participants will be recruited (could be anyone from our families, classmates, housemates or even random people we meet on the streets).

- Demographic Survey: Prior to the negotiation, participants complete a brief, optional survey capturing age, gender, self-assessed negotiation experience (Likert scale), and native language. This data allows for post-hoc analysis of potential interactional biases.
- Informed Consent: Participants are provided with a clear information sheet and must give explicit consent, acknowledging the study's purpose, data usage, and anonymity guarantees.

3.3 Technical Implementation

3.3.1 Platform Architecture

A full-stack web application is built to facilitate the experiments.

- Backend Framework: Django (Python) handles server-side logic, user authentication, session management, and API orchestration.
- Database: PostgreSQL stores all structured data, including:
    - UserProfile: Demographic data.
    - NegotiationSession: Session ID, final outcome, profit
    - DialogueTurn: Sequential record of each message (Human/AI, content, timestamp)
    - OfferHistory: Track all numerical offers proposed by the human and AI.
- Frontend Interface: A clean, real-time chat interface is built using React to simulate a fluid conversational experience.

3.3.2 AI Negotiator

The AI counterpart is implemented as a rigorously prompt-engineered LLM.

- Model: GPT-4 (OpenAI API) serves as the core reasoning engine.
- System Prompt Engineering: The agent's behavior is controlled via a structured system prompt that defines:
    - Role & Goal: Clearly states its assigned role (Seller) and RP.
    - Strategy Instructions: Directed to be "firm but reasonable," make gradual concessions, and justify its positions.
    - Structured Output Format: Required to respond in a JSON schema separating {“reasoning”: ”...”, “message”: ”...”, “interpreted_offer”: $} to ensure parsability.
    - Decision Logic: Explicit rules for the final decision (e.g., "Accept if the final offer is within 5% of your RP; otherwise, decline").

- State Management: The backend maintains the conversation history and injects it into each API call to provide the LLM with context.

3.4 Data Collection and Variables

3.4.1 Primary Data Collected

- Dependent Variables (Outcomes):
  - Agreement Rate: Proportion of sessions ending in a deal.
  - Final Price: The agreed upon price (if deal) or last offer (if no deal).
  - Individual Profit: For each party, calculated as |Final Price - RP|.
  - Joint Gain: Sum of profits for both parties (maximized at the midpoint).
- Independent Variables (Conditions):
  - Participant Demographics: Age group, gender, experience level.
- Process Variables:
  - Concession Pattern: Rate and size of numerical offers change.
  - Conversation Content: Full textual transcript for linguistic analysis.
  - Turn Count: Up to 5 turns per session.

3.5 Analytical Plan

3.5.1 Quantitative Analysis

- Descriptive Statistics: Calculate means, standard deviations, and frequencies for all outcome variables (agreement rate, final price, profit).
- Comparative Tests:
  - T-tests/ANOVA: Compare average final prices and profits by participant role and demographic subgroups to detect outcome disparities.
  - Chi-square tests: Examine if agreement rates differ significantly based on human demographics.
- Correlation and Regression: Model Final Price or Profit as a function of demographic factors and concession patterns to identify predictive variables.

3.5.2 Qualitative and Linguistic Analysis

- Thematic Analysis: Transcripts will be coded for recurring negotiation strategies (e.g., persuasion tactics, fairness appeals, threats, rapport-building).
- Linguistic Inquiry: Using LIWC (Linguistic Inquiry and Word Count) or similar tools, analyze the AI's language for emotional tone, formality, and assertiveness across sessions with different human counterparts.

- Bias Detection: Manually review edge-case interactions (e.g., all declined offers for a specific demographic) to identify potential patterns of unfair or inconsistent AI behavior not captured quantitatively.

3.5.3 Validation and Reliability

- Pilot Testing: A pilot with 10-15 participants will be conducted to refine prompts, interface usability, and experimental timing.
- AI Consistency Check: A subset of identical human offer sequences will be presented to the AI multiple times to verify the stability and determinism of its responses.
- Inter-coder Reliability: For qualitative coding, two researchers will independently code a sample of transcripts, with Cohen's Kappa calculated to ensure reliability before full scale analysis.

3.6 Ethical Considerations

- Anonymity: All personal identifiers will be removed from the dataset. Participants will be referenced by unique session IDs only in analysis and reporting.
- Data Security: Data will be stored on a secure, password protected server with encrypted connections. Raw data will be retained only for the duration of the analysis.
- Debriefing: Upon completion, participants will receive a debriefing form explaining the study's full purpose and hypotheses.

This methodology provides a replicable, multi faceted approach to capturing and analyzing the dynamics and potential biases inherent in human-AI negotiation.

4. Project Management

4.1 Project Organization

4.1.1 Project supervision

The project is supervised by Mr Samuel Deleplanque, lecturer at JUNIA, and Mr Luis Fernando Perez Armas, lecturer at IESEG.

Supervision meetings are held to monitor progress, validate intermediate deliverables, and guide the next steps of the project.

Each meeting is documented through written minutes to ensure traceability of decisions and feedback.

4.1.2 Team composition

The project is carried out by a team of four students, with clearly defined roles to ensure efficient organization:

- **Nisy : Project Lead & AI Integrator**
  Responsible for overall project coordination and the integration of AI-related components (LLM API integration).
  She also acts as the main point of contact with the supervisors and ensures global coherence.

- **Antoine : Experimental Protocol Lead (Front-end)**
  Responsible for defining and managing the experimental protocol, including the implementation of user interactions on the front-end (UI/UX).
  He ensures consistency and reproducibility of the experiments.

- **Clarence : Technical Lead (Back-end)**
  Responsible for back-end development, API integration, and system stability.
  He ensures the technical reliability of the experimental framework.

- **Tetiana : Data Analysis Lead (Database)**
  Responsible for data structuring, storage, and analysis.
  She oversees the database and prepares quantitative and qualitative analyses of the results.

4.2 Planning and Milestones (Semester 1)

Semester 1 is dedicated to project scoping, literature review, and methodological design.
The project planning follows a structured four phase organization, in accordance with the official project schedule provided by the institution.

- **Week 1 – Project launch and scoping**
  This phase focuses on understanding the project topic, its context, and its objectives.
  The team establishes the project organization, defines roles and responsibilities, and sets up the working tools and initial planning.
  Kick-off and follow-up meetings with the supervisors are organized to validate the project scope.

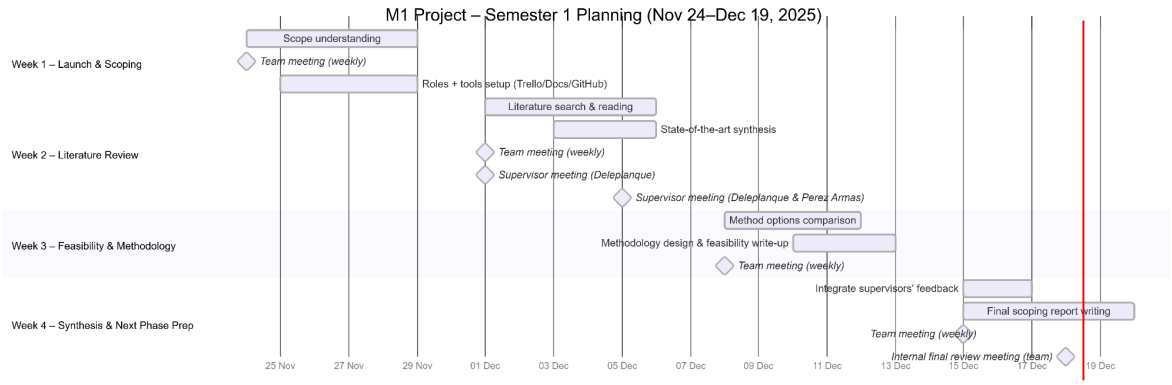
- **Week 2 – Literature review and state of the art**
  This phase is dedicated to conducting a structured literature review related to negotiation, biases, and generative artificial intelligence.
  The team analyzes existing research, identifies key approaches and limitations, and produces a written synthesis of the state of the art.
  Progress and validation meetings with the supervisors allow refinement of the research direction.

- **Week 3 – Feasibility study and methodological design**
  During this phase, the team evaluates different methodological options and assesses the technical and scientific feasibility of the proposed approach.
  The selected methodology is validated through meetings with the supervisors.

- **Week 4 – Synthesis and preparation for the next phase**

  This final phase of Semester 1 focuses on consolidating all preparatory work.
  Feedback from the supervisors is integrated, and the final scoping report is completed, including the project context, state of the art, methodology, and feasibility analysis.
  This phase also prepares the transition toward the experimental phase planned for Semester 2.

The overall planning and progression of these phases are summarized using a Gantt-style schedule, highlighting key milestones and associated deliverables.



4.3 Work Methods and Tools

To ensure efficient collaboration and structured project management, several tools are used throughout the project:

- **Microsoft Teams** is used for internal communication, document sharing, and the organization of online meetings with supervisors and team members.
- **Trello** is used to manage tasks, assign responsibilities, and track project progress according to defined responsibilities.
- **Google Docs** is used for collaborative writing and sharing of project documents, enabling real-time contributions from all team members.
- **GitHub** is used for version control and collaborative development of the project's codebase, ensuring traceability of technical contributions and changes.
- **Google Scholar** is used to conduct academic literature searches and identify relevant scientific publications related to negotiation, biases, and generative artificial intelligence.
- **Canva** is used to create presentation supports, ensuring clear and professional visual communication of the project content.

These tools support clear communication, effective coordination, and transparency in task progression throughout the project lifecycle.

5. Conclusion

The state of the art confirms that bias in negotiation with Generative AI is a real, measurable, and multifaceted phenomenon. LLMs do not negotiate as neutral optimizers: they embed cognitive heuristics and societal prejudices into their strategic outputs. While methodological rigor is increasing and mitigation strategies are being explored, the field is nascent. The prevailing consensus warns against the uncritical, instrumental deployment of AI in sensitive negotiation contexts. Future research must move beyond documenting bias to developing robust, fair, and culturally-aware AI negotiation systems, while critically examining the human and relational changes these technologies instigate. The central challenge remains: designing AI that enhances, rather than inequitably distorts, the fundamentally human process of negotiation.

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Appendices

J'ai fait jouer des IA au Monopoly… elles ont craqué.

Example of AI negotiation prompt

Negotiating with AI: The future of customer interactions online

Why AI Gives Negotiators 48% Better Outcomes

Zeuthen strategy

Computers arguing with people

Appendix

Pharmaceutical Plant Negotiation Details

General information (common to both negotiators):

1. Plant purchased by current owner 2 years ago for $15 million. The seller at that time was facing bankruptcy.
2. Plant appraised for $19 million 2 years ago.
3. Similar plant, although newer, sold for $26 million 9 months ago.

Buyer information:

1. Best alternative to this negotiated agreement: $25 million it would cost to build a new plant. Given perfect conditions, the plant would take at least a year to be fully operational with Food and Drug Administration approval.

Seller information:

1. Best alternative to this negotiated agreement: $17 million from stripping plant and selling its parts and equipment separately.